

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination April/May 2016

PH819.09 PHARMACEUTICAL PRODUCTION MANAGEMENT, TECHNOLOGY AND DRUG REGULATORY AFFAIRS

Date: 09/05/2016, Monday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Write a note on plant design for capsule manufacturing unit. 05
B) Explain in-process quality control (IPQC) tests for tablet. 05
- Q.2 A) Explain the mechanical hazards and its prevention. 05
B) Explain in-process quality control (IPQC) tests for emulsion. 05

OR

- Q.2 A) What is pharmaceutical waste? Describe the waste disposal procedures. 05
B) Explain in brief about filling of Investigational New Drug Application (INDA). 05
- Q.3 Answer the following (Any three) 15
(A) Enlist equipments required in manufacturing of solid, liquid and parenteral dosage forms at pilot level.
(B) Differentiate between NDA and ANDA
(C) Explain manufacturing and in-process quality control (IPQC) tests for soft gelatin capsule.
(D) What are the requirements of product registration in India?

SECTION-II

- Q.4 A) Write a note on "Schedule Y" of D and C Act. 05
B) Explain the process flow and layout for the tablet manufacturing unit. 05
- Q.5 A) What is technology transfer? Write its importance in scale-up process. 05
B) Explain in-process quality control (IPQC) tests for small volume parenterals. 05

OR

- Q.5 A) Explain the importance of production planning. Discuss in brief the suitable methods for inventory control. 05
B) Explain in brief WHO as a certification organization. 05
- Q.6 Answer the following (Any three) 15
(A) Explain in brief scale up for liquid oral manufacturing.
(B) Explain the objectives of scale up process.
(C) Draw and explain flow plan for manufacturing of large volume parenteral products.
(D) Explain about SUPAC guidance on immediate release solid oral dosage forms.